

SOLAR ROOFTOP SYSTEMS
A DIY GUIDE

₹50,000
Cost of an RTS installation for a 1 kW (kilowatt) system

4-5.5 Units
Electricity generated by a 1 kW solar power plant on a clear, sunny day

₹30,000
Government subsidy under Muft Bijli Yojana for 1 kW system (Consumer pays Rs 20,000). For 2kW it's Rs 60,000 (consumer share Rs 40,000) and Rs 78,000 for 3 kW (consumer share Rs 66,999)

25 Years
Expected life of solar PV modules. For maintenance, periodic cleaning of solar panels is required

5.32 Years
Time taken to recover costs for a 3 kW RTS system through savings on power bills and sale of excess power to discoms, assuming average electricity charges of Rs 2,000 per month

Source: https://www.pmsuryaghar.gov.in/

For the past two years since Bengaluru resident C.M. Dinakar installed a 4 kilowatt (kW) solar power plant on the roof of his house in the city's Basaveshwara Nagar locality, he hasn't had to worry about his electricity bill. In fact, he's using more power these days. Dinakar's rooftop solar system, which cost him Rs 3 lakh, generates an average of 16 units of electricity a day—more than what his household consumes daily. The excess is sold to the power grid. Even though Karnataka introduced a free electricity scheme for households last year, he's happy with the heavy upfront investment he made on solar photovoltaic (or solar cell) panels to generate his own power. It's better than putting money in a fixed deposit, feels Dinakar, an electrical engineer.

Till recently, people like Dinakar were part of a small tribe. This is because rooftop solar (RTS) hadn't caught on in India. In 2014, the government had initiated a Rooftop Solar Programme, which aimed at generating 40 gigawatts (GW) from rooftops by 2022. The target proved elusive—by November 2023, the cumulative installed capacity was estimated at 10.4 GW, with 2.65 GW on residential rooftops, the rest from large-scale institutional arrays. The 2014 programme faltered due to lack of awareness, high initial costs, issues with grid stability and limited skilled workforce. With about half a million residential RTS installations across India at the beginning of this year according to industry estimates, a course correction was needed.

That very need was fulfilled through the renewed push for RTS through the ambitious 'PM-Surya Ghar: Muft Bijli Yojana' launched in February with the goal of installing RTS in 10 million households within three years through a subsidy outlay of Rs 75,021 crore. On March 16, Prime Minister Narendra Modi announced that over 10 million households had signed up for the scheme. "Registrations have been pouring in...Assam, Bihar, Gujarat, Maharashtra, Odisha, Tamil Nadu and Uttar Pradesh have seen over 5 lakh registrations," the PM said on X. There were 12.8 million registrations and 1.4 million applications as of July 23. According to the Union ministry of new and renewable energy (MNRE), the total installed capacity of residential RTS by end-July stood at 4 GW—an increase of 1.35 GW in eight months—with Gujarat, Maharashtra, Kerala, Uttar Pradesh and Rajasthan making up the top five states.

Indian households consumed about a fourth of the 1,317 billion units of electricity sold in the country by 2022. If we have to meet the renewable energy target of 500 GW installed capacity by 2030 (it's currently at 203 GW), the adoption of solar power, including RTS, will have to be accelerated. Furthermore, once the cost of installing RTS—aided by a generous subsidy under the Muft Bijli Yojana—has been recouped through savings on power bills and sale of excess electricity to a discom, households can enjoy free power for a 25-year period which is the normal lifetime of the solar panels. At the current pace, the country will likely reach 15 GW of RTS capacity cumulatively across industrial, commercial and res-